

Interactive Extinction Risk Page — Complete Design & Scientific Content Guide

This document provides a highly detailed breakdown for the “Extinction Risk” section of the Hawaiian Coot endangered species website project.

The goal is to transform a static science page into a cinematic, interactive educational experience inspired by museum exhibits, conservation storytelling, and documentary design.

Overall Purpose of the Page

The Extinction Risk page explains why the Hawaiian Coot remains vulnerable despite conservation progress.

The page should emotionally communicate urgency while also showing hope through conservation efforts.

Key Goals:

- Educate visitors about extinction threats.
- Explain ecological pressures.
- Visualize population vulnerability.
- Show conservation success.
- Encourage environmental awareness.

Desired emotional tone:

- Serious
 - Scientific
 - Cinematic
 - Hopeful
-

Visual Theme and Color Palette

Recommended Colors:

- Deep black backgrounds
- Crimson red for danger indicators
- Gold for headings
- Dark green and teal for ecosystem elements

The page should feel:

- Elegant
- Immersive
- Museum-like
- Modern

Avoid:

- Neon colors
 - Cartoon styling
 - Excessively bright themes
-

Interactive Threat Matrix

The threat bars should animate dynamically.

Interactions:

- Bars grow when scrolled into view.
- Hovering reveals detailed scientific explanations.
- Clicking opens expanded information panels.

Threat categories:

1. Habitat Loss
2. Invasive Predators
3. Climate Change
4. Disease
5. Human Disturbance

Expanded content examples:

Habitat Loss:

- Wetland drainage
- Urban development
- Loss of nesting habitat

Invasive Predators:

- Mongooses
 - Rats
 - Feral cats
-

Population Visualization System

The population dot matrix should become interactive.

Features:

- Each dot represents 10 birds.
- Hover reveals population groupings.
- Dots disappear during disaster simulations.
- Conservation recovery rebuilds the population grid.

Educational Purpose:

Users visually understand how quickly populations can decline.

Environmental Simulation System

Suggested simulation modes:

- Present Day
- Severe Drought
- Hurricane Season
- Conservation Success

Simulation effects:

- Water level changes

- Population updates
 - Threat indicators intensify
 - Wetland vegetation changes
-

Cinematic Scroll Effects

Recommended animation effects:

- Slow fade-in titles
- Sequential threat animations
- Building dot matrices
- Water ripple backgrounds
- Subtle drifting mist

Goal:

Create a documentary-style interactive experience.

Scientific Information to Include

Suggested data:

- Current Hawaiian Coot population
- Historical population decline
- Wetland restoration acreage
- Predator control statistics
- Climate vulnerability indicators

Important concepts:

- Biodiversity
 - Habitat fragmentation
 - Ecological resilience
 - Wetland conservation
-

Accessibility and User Experience

Accessibility recommendations:

- High contrast text
- Readable typography
- Motion reduction settings
- Keyboard navigation
- Mobile optimization

Mobile layouts should:

- Stack content vertically
 - Preserve readability
 - Reduce animation intensity
-

Typography Recommendations

Suggested font usage:

- Elegant serif for major titles
- Clean sans-serif for body text
- Minimal cursive for decorative emphasis

Avoid:

- Too many font families
 - Excessive decorative fonts
-

Final Creative Direction

The finished page should feel like a professional conservation exhibit.

Visitors should leave feeling:

- Concern for endangered species
- Appreciation for wetlands

- Hope in conservation efforts

Core message:

“The species remains vulnerable, but conservation can make a difference.”
